

■ Sports Marmalade Energy Bites – 2:1 Glucose:Fructose (HM Pectin + Concentrated Juice 1:4)

Approx. 40 bites, each 10.5 g ($\approx$ 0.37 oz) – total yield $\approx$ 455 g (1 lb) + 60 g waste

■ Carbohydrate Base

Ingredient	Amount (g / oz)	Comment
Sugar (granulated)	100 g / 3.5 oz	Base, helps pectin to gel
Glucose syrup (light)	140 g / 4.9 oz	Adds elasticity and stable energy release
Maltodextrin	80 g / 2.8 oz	Reduces sweetness, adds stability
Fructose	80 g / 2.8 oz	Balances to 2:1 glucose:fructose ratio
Conc. juice 1:4	50 g / 1.8 oz	Provides flavour and acidity
Water	130 g / 4.6 oz	Total liquid balance $\approx$ 180 g (6.3 oz)

■ Gelling Agent & Acid

Ingredient	Amount (g / oz)	Comment
Pectin (HM)	20 g / 0.7 oz	High methoxyl pectin – not Melatin
Citric acid	1.5 g / 0.05 oz	Activates pectin – add after boiling

■ Electrolytes (optional but recommended)

Ingredient	Amount (g / oz)	Comment
Electrolyte tablets (3 pcs)	12 g / 0.4 oz	Dissolve in water before heating
Alternative: sea salt + KCl + magnesium citrate	—	1 g + 0.5 g + 0.3 g

■ Instructions

1. Combine water, concentrated juice, and (if used) electrolytes in a saucepan.
2. Whisk in the pectin and let rest for 1–2 minutes to hydrate.
3. Add sugar, glucose syrup, fructose, and maltodextrin.
4. Boil while stirring until temperature reaches 106–107 °C (223–225 °F) – large, slow bubbles should appear.
5. Remove from heat.
6. Quickly stir in citric acid.
7. Pour the mass into silicone moulds (about 6.5 g / 0.23 oz per cavity) or onto parchment paper.
8. Let sit 1 hour at room temperature, then refrigerate for 3–4 hours.
9. Unmould and dust with a 50/50 mix of potato and corn starch to prevent stickiness.

■ Tips

- Stir continuously while hydrating pectin – clumps will ruin the gel.
- Do not exceed 108 °C (226 °F) or the gel will become chewy.
- For a lighter flavour, reduce concentrated juice to 40 g (1.4 oz) and increase water to 140 g (4.9 oz).

■ **Nutrition (per bite $\approx$ 10.5 g / 0.37 oz)**

Nutrient	Amount	Comment
Carbohydrates	8.7 g (0.31 oz)	$\approx$ 35 kcal
Energy	35 kcal	—
Sodium	$\approx$ 70 mg	If electrolytes are included
Potassium	$\approx$ 45 mg	—
Magnesium	$\approx$ 12 mg	—